Spirit Box: Week 4 plan

Hacktoberfest Open-Source AI Challenge Week 4, "Ghost in the Machine". Whisper hears ghosts in room noise; Gemma decides which voices are real. Lives in `Week4/` of Phantom8420/Hacktoberfest.

Phase 0: Kill test

- Install faster-whisper (already used by jarvis). Record a few minutes each of silence, room tone, fan, and rain / pink noise.
- Transcribe each with tiny, base and small, once with Whisper's guards on (VAD, no-speech threshold, log-prob cutoff) and once with them off.
- Smoke-test transformers.js whisper-tiny in a browser on the same noise.

> **Gate.** With guards off, do we get a steady stream of ghosts per minute? Yes: go on. Only on recorded noise: make noise the spirit-box input. Almost none: drop the idea now, before anything is built. The browser result decides Phase 5's shape.

Phase 1: Summon

- `python -m spiritbox summon`: record or load audio, cut it into windows, run Whisper with guards off.
- Every contact goes to a JSONL log: text, time, model, noise type, and Whisper's own numbers (no-speech probability, average log-prob, compression ratio).
- Small tests for windowing and logging.

> **Done when** one command turns a recording into a log of contacts.

Phase 2: Witness (Gemma)

- Reinstall Ollama and a Gemma 3 model. Gemma gets each segment plus Whisper's numbers and answers in a fixed JSON schema: real speech or ghost, and the kind (sign-off, captions credit, [Music]-style tag, repetition loop, wrong language, other).
- Build a labelled set: ghosts from Phase 1 plus real speech (me reading text aloud, a few public-domain clips).
- Measure Gemma against a no-model baseline (Whisper's own thresholds plus a list of known ghost phrases). If the baseline wins, the post says so.

> **Done when** there are precision and recall numbers for Gemma and for the baseline.

Phase 3: Census

- Run every noise type with every model size, each recording twice.
- Answer: which noise summons the most, do bigger models haunt less or differently, the top ghosts, whether the same noise brings back the same ghost, and how often the language drifts.
- Charts. Offensive ghosts are filtered from anything shown and only counted.

> **Done when** the numbers and charts the post will quote exist and come from a script.

Phase 4: Exorcise

- Paste a transcript, or give an audio file, and get back the lines that are probably hallucinated, with the reason.
- Run it on a real Whisper transcript with planted silence to show it catches the phantom lines in practice.

> **Done when** the tool is useful outside the joke: it cleans a real transcript.

Phase 5: Séance page (UI and demo)

- Local page: record, watch contacts appear live, Gemma's verdict on each, census charts, the exorcise box.
- Online demo: whisper-tiny in the browser on the judge's own mic, if Phase 0 said it works. Gemma can't run there, so labels come from the baseline and the page says so. Deploy to Vercel.
- Fallback: a narrated screen recording.
- Design skills first, then palette and fonts chosen for this project. Checked at desktop and mobile widths, dark and light.

> **Done when** a judge can summon their own ghost, or watch a video of it, without installing anything.

Phase 6: Write-up

- README with how to run it, the measurements and the limits.
- DEV post draft (`published: false`), cover image, screenshots, demo.
- Re-read and cite the "Careless Whisper" paper.
- The open-weights / closed-data paragraph: the ghosts' origin is a hypothesis, not a proof.
- Truthful AI disclosure. Commit and push every step.

> **Done when** the draft is ready for me to publish.

If time runs short, cut in this order

1. Ghost voice read aloud (Piper).
2. Census breadth: fewer noise types.
3. Browser demo becomes the video.
4. Exorcise for audio files only.

Never cut: the kill test, Gemma's evaluation against the baseline, and the write-up.